SCG-FSOD: Semantic Correlation-Guided Few-Shot Object Detection in Remote Sensing Images

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***Abstract*—Few-shot object detection (FSOD) in remote sensing imagery aims to achieve accurate detection of novel object categories with limited training samples. However, current mainstream transfer learning-based methods are frequently constrained by two major challenges. First, due to the scarcity of novel class samples and the bias of pretrained models toward base classes, novel classes may rely overly on base class knowledge to construct feature representations, causing novel class features to be easily confused with similar base classes. Second, the interclass similarity and intraclass diversity in remote sensing images can further exacerbate classification confusion. To tackle the above problems, we propose a semantic correlation-guided FSOD (SCG-FSOD) in remote sensing images. Specifically, we design an interclass semantic correlation transfer (ISCT) module to fully explore the semantic correlations between base and novel classes, employing knowledge distillation for cross-class correlation transfer. This module effectively mitigates base-class bias while enhancing the discriminative power of novel class features. Furthermore, we propose a semantic correlation-driven supervised contrastive learning (SSCL) module, which employs semantic correlation priors to weight negative sample pairs in supervised contrastive learning. By imposing stronger separation constraints on negative pairs with high interclass similarity, this module significantly alleviates feature confusion in remote sensing images. Extensive experiments conducted on two public benchmark datasets (DIOR and NWPU VHR-10.v2) demonstrate the effectiveness of our proposed method, which achieves competitive performance compared with several state-of-the-art approaches.**

***Index Terms*—Few-shot object detection (FSOD), knowledge distillation, semantic correlation.**

I. Introduction

In recent years, object detection in remote sensing images [1], [2], [3], [4], [5] has attracted widespread attention, aiming to localize and recognize regions of interest within the imagery. As a fundamental task in remote sensing image interpretation, object detection has been widely applied in various fields, such as environmental monitoring, disaster prediction, and military reconnaissance [6], [7]. Benefiting from the development of deep learning, particularly deep convolutional neural networks [8], [9], [10], and transformers [11], [12], [13], object detection has achieved remarkable progress. Although current object detection algorithms achieve satisfactory performance, their training process demands a substantial amount of annotated samples. However, annotating remote sensing images demands expert knowledge and substantial time investment, while sufficient samples for rare categories are often difficult to acquire. To reduce the reliance on annotated samples, few-shot learning [14], [15], [16], [17], [18], [19] has attracted substantial research interest, as it enables models to achieve promising generalization performance even under label-scarce conditions. As an important extension of few-shot learning, few-shot object detection (FSOD) has drawn widespread attention.

FSOD aims to leverage knowledge learned from base classes with abundant annotated samples to facilitate the detection of novel categories that have only limited instances. In recent years, researchers have developed numerous successful FSOD approaches, which can be broadly categorized into two paradigms: meta-learning-based methods [20], [21], [22], [23], [24], [25] and transfer learning-based methods [26], [27], [28], [29], [30], [31]. Meta-learning approaches train models through episodic meta-training on base classes to simulate few-shot learning tasks, enabling rapid adaptation to novel categories. However, meta-learning methods suffer from complex data organization and low training efficiency, which limits their application scenarios.

To achieve simple and efficient training, many existing FSOD methods adopt the transfer learning paradigm. This paradigm initially learns transferable feature representations on base classes with abundant annotations and subsequently adapts the model to novel classes with limited annotations through fine-tuning. For example, Zhu et al. [32] proposed a novel FSOD method called RIFR for remote sensing images, which employs a fine-tuning training paradigm to tackle large intraclass scale variations through multiscale ROI features while introducing a PT loss function to alleviate classification confusion. Benefiting from its streamlined data organization and computational efficiency, the transfer learning paradigm has garnered significant research attention in the remote sensing community. Although transfer learning-based methods have achieved considerable success, their performance remains

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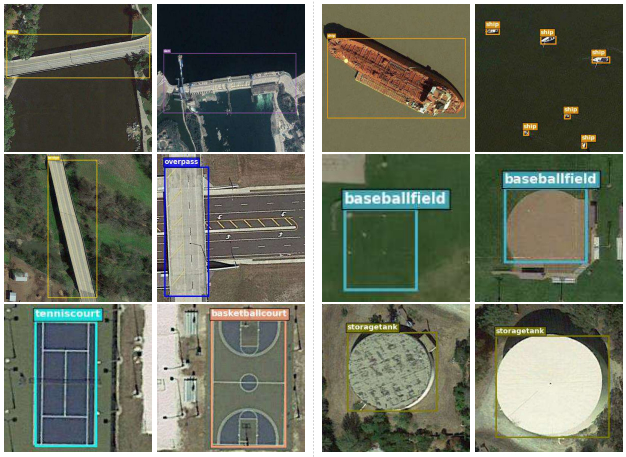
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(a) (b)

Fig. 1. Demonstration of (a) interclass similarity and (b) intraclass diversity in remote sensing images.

constrained by a fundamental challenge. Due to the limited number of novel class samples and the inherent bias of pretrained models toward base classes, novel classes may rely excessively on base class knowledge to construct their feature representations. This reliance can lead to confusion between novel-class features and similar base categories, consequently compromising the model's performance. This fundamental issue is further exacerbated in remote sensing scenarios. The inherent characteristics of remote sensing images, characterized by high interclass similarity and significant intraclass diversity, intensify the feature confusion problem. For instance, from a bird's-eye perspective, certain object categories (e.g., "bridge" and "dam," "bridge" and "overpass," "tennis court" and "basketball court") may exhibit high visual similarity, while objects within the same category (e.g., "ship," "baseball field," and "storage tank") can display considerable appearance variations, as illustrated in Fig. 1. Existing methods struggle to learn sufficiently robust discriminative features from limited novel-class samples during the fine-tuning stage, which often leads to the misclassification of novel-class objects as visually similar base classes.

To address the above issues, we propose a semantic correlation-guided FSOD (SCG-FSOD) method for remote sensing images. Based on the fact that the semantic relationships between novel and base classes remain constant [33], [34], our method incorporates the inherent semantic relationships among image categories as auxiliary information to enhance detection performance on novel classes. Specifically, our method innovatively designs two key components: the interclass semantic correlation transfer (ISCT) module and the semantic correlation-driven supervised contrastive learning (SSCL) module. The ISCT module first designs a memory bank update strategy and an adaptive prototype calibration module to generate accurate class prototypes for each object category. Furthermore, we employ the ProtoNegative loss function to enhance the discriminability of each class prototype. Subsequently, we utilize the similarity between ROI features and class prototypes as soft labels to guide the detector in learning interclass semantic correlations through knowledge distillation, thereby mitigating base-class bias and enhancing the discriminative power of novel class features. However, to address the issue of high interclass similarity and large intraclass variance in remote sensing images exacerbates classification confusion, we propose the SSCL module. This module employs prior knowledge to weight negative pairs in supervised contrastive learning, enforcing stronger separation constraints on negative pairs of highly similar categories in the feature space. Under such constraints, the detector achieves improved distinction between objects of different categories, thus effectively alleviating the problem of classification confusion. Extensive experiments conducted on both DIOR [35] and NWPU VHR-10.v2 [36] datasets demonstrate the effectiveness of our method, achieving competitive performance compared to several state-of-the-art approaches.

In summary, the main contributions of this work can be summarized as follows.

1) We propose a novel SCG-FSOD framework that innovatively introduces interclass semantic correlation into remote sensing FSOD tasks, effectively enhancing the model's detection performance for novel objects.
2) We design an ISCT module that thoroughly explores semantic correlations between base and novel classes and incorporates knowledge distillation to guide the detector in learning interclass semantic correlation, thereby effectively mitigating base-class bias and enhancing the discriminability of novel class features.
3) We propose a SSCL module that leverages semantic prior knowledge to guide the contrastive learning process, thereby enhancing the feature discriminability of models in remote sensing scenarios.
4) Extensive experiments on two benchmark remote sensing datasets, DIOR and NWPU VHR-10.v2, demonstrate that our method significantly outperforms multiple state-of-the-art baseline models.

The rest of this article is organized as follows. Section II reviews the related work, including FSOD, FSOD in remote sensing images, and contrastive learning. Section III introduces the problem formulation of FSOD and our proposed SCG-FSOD method. Section IV presents the experimental results on two public remote sensing FSOD datasets, demonstrating the effectiveness of our proposed SCG-FSOD method. Finally, we draw conclusions in Section V.

II. Related Works

A. Few-Shot Object Detection

FSOD aims to detect novel object categories with only a few annotated examples by leveraging prior knowledge from base classes. Existing approaches can be broadly categorized into meta-learning-based methods and transfer learning-based methods. Meta-learning methods simulate few-shot scenarios during training, enabling models to generalize to novel classes via episodic learning. For instance, the early work Meta R-CNN [20] introduced feature reweighting to dynamically align support set features with query images, establishing a foundation for subsequent research. Subsequently, Fan et al.

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[37] recognized the importance of suppressing background interference and integrated the attention mechanism into region proposal generation, while simultaneously optimizing feature alignment through multirelation matching. In recent years, with the rise of Transformer architectures, Meta-DETR [23] integrates the DETR framework with meta-learning and proposes a correlational aggregation module that enables efficient feature interaction between support and query sets through feature matching and task encoding, thereby significantly improving the generalization capability for FSOD. Although meta-learning approaches have achieved remarkable progress in FSOD, they suffer from complex data organization and low training efficiency, significantly limiting their practical applications.

In contrast to meta-learning approaches, transfer learning-based methods first pretrain the detector on base classes and then adapt it to novel classes through limited supervised fine-tuning, thereby simplifying few-shot detection. The representative TFA framework [26] achieves strong performance through a simple approach involving frozen backbone networks with fine-tuning limited to the classifier and bounding box regressor. FSCE [27] enhances feature discriminability through contrastive learning integration, thereby improving classification performance. Meanwhile, DeFRCN [28] mitigates multistage contradictions and multitask conflicts through gradient decoupling and prototype calibration. Fan et al. [38] proposed a model calibration framework comprising backbone calibration, uncertainty-aware RPN, and detector calibration, which systematically addresses model bias issues in FSOD. DiGeo [39] introduced the geometry-aware discriminative learning that enhances feature discriminability for novel classes through spatial contrastive objectives and covariance constraints. In contrast, SMILe [30] innovatively employs a submodular mutual information learning framework, constructing more distinctive feature clusters via combinatorial mutual information functions. These approaches have collectively advanced the development of transfer learning paradigms in FSOD, progressively addressing the core challenges posed by data scarcity.

Despite remarkable advances in transfer learning approaches, the scarcity of novel class samples and the preference of pretrained models for base classes, novel classes may overly rely on base class knowledge to construct feature representations. This often results in ambiguous discrimination between novel classes and their similar base counterparts. To address this, we propose the interclass semantic correlation transfer module, which systematically explores semantic correlations between base and novel classes while incorporating knowledge distillation techniques to enable more discriminative feature learning for novel categories.

B. FSOD in Remote Sensing Images

Compared with natural images, remote sensing images are typically captured by satellite or aerial sensors in a vertical top-down view, containing complex background information. However, they exhibit large intraclass variations and high interclass similarities, which significantly increase the difficulty of FSOD in remote sensing scenarios. To address this challenge, researchers have conducted extensive research on FSOD in remote sensing images and developed numerous effective approaches. For instance, P-CNN [40] employs class-aware prototypes and rotation-enhanced feature learning to address the FSOD problem in remote sensing images, thereby enhancing robustness against orientation variations and background clutter. G-FSDet [41] employs metric discrimination loss and representation compensation to boost novel-class detection while mitigating base-class forgetting. Li et al. [42] proposed a FSOD method that combines deep metric learning and knowledge inheritance. By jointly optimizing multisimilarity loss and coherence loss, it can effectively detect novel categories in aerial images while maintaining the performance of base categories. RP-FSOD [43] incorporates a retentive compensation mechanism and a personality feature filtering strategy to improve feature discriminability. GE-FSOD [44] developed the cross-level feature fusion pyramid attention network, the multistage refined region proposal network, and the generalized classification loss to enhance the model's generalization capacity. GIDR [45] enhances foreground features through global support information and achieves adaptive feature alignment by establishing relative position correspondence between query and support features, significantly improving the model's robustness against object orientation variations and background clutter.

Despite the promising progress achieved in existing studies, intraclass variation and interclass similarity in remote sensing images remain critical challenges for FSOD. The scarcity of novel-class samples further exacerbates the difficulty in learning discriminative features. To address these issues, we propose an SSCL module. This module simultaneously pulls positive sample pairs closer and pushes negative sample pairs apart in the feature space, thereby effectively minimizing intraclass variance while maximizing interclass variance.

C. Contrastive Learning

Contrastive learning has emerged as a powerful paradigm in representation learning [46], [47], [48], [49], [50], aiming to learn meaningful features by contrasting positive and negative samples in an embedding space. The core idea is to pull semantically similar instances (positives) closer while pushing dissimilar ones (negatives) apart. Early self-supervised contrastive learning methods, such as SimCLR [49], demonstrated that simple instance discrimination tasks with strong data augmentation could yield competitive representations. MoCo [50] further improved scalability by introducing a dynamic memory bank and momentum encoder to maintain a large set of negatives. BYOL [51] eliminated the need for negative samples by relying on asymmetric network updates and a predictor head. These methods have shown that contrastive learning can rival supervised learning in downstream tasks, particularly when labeled data is scarce.

Based on the above insights, researchers began to explore how to further enhance contrastive learning by incorporating supervised signals. This effort has led to the development of supervised contrastive learning methods, which leverage

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label information to construct more semantically meaningful positive and negative sample pairs. The core idea is to treat samples from the same class as positive pairs and those from different classes as negative pairs, thereby optimizing intraclass compactness and interclass separation. Khosla et al. [52] formalized this approach as the supervised contrastive loss, which generalizes traditional cross-entropy by incorporating multiple positive samples per anchor. Empirical results demonstrate that this method enhances model robustness and generalization capability.

In supervised contrastive learning, all negative sample pairs typically carry equal weights. However, the actual dissimilarity among negative pairs can vary significantly. To address this, we introduce semantic correlation between categories to weight negative sample pairs in supervised contrastive learning, thereby enhancing the discriminative power of the learned features.

III. Methodology

A. Problem Formulation

In this article, we follow the existing FSOD settings [40], [41], [53] in remote sensing images. All classes in the dataset are divided into two parts: base class C_{base} and novel class C_{novel}, where $C_{\text{base}} \cap C_{\text{novel}} = \emptyset$. Concretely, each category in C_{base} contains abundant annotated samples, while each category in C_{novel} is limited to only K annotated instances (e.g., $K = 3, 5, 10, 20$). The base dataset D_{base} contains all annotated samples from C_{base}, formally denoted as $D_{\text{base}} = \{(x_i, y_i), y_i \in C_{\text{base}}\}$. The novel dataset D_{novel} is composed of K annotated samples from C_{novel}, represented as $D_{\text{novel}} = \{(x_i, y_i), y_i \in C_{\text{novel}}\}$, where x_i represents the image, and y_i denotes the corresponding label. Based on the above settings, our model training adopts a two-stage fine-tuning paradigm. During the base training phase, the base model is trained using the extensively annotated dataset D_{base}. Subsequently, in the few-shot fine-tuning phase, we construct a balanced training set by uniformly sampling K instances per class from both D_{base} and D_{novel}, followed by transferring and fine-tuning the pretrained model's knowledge on this balanced set to enhance its generalization capability in data-limited scenarios.

B. Overall Framework

The proposed SCG-FSOD method is implemented based on Faster R-CNN [54], which incorporates the ISCT and SSCL modules, as illustrated in Fig. 2. Specifically, the ROI features are extracted from the input image using the ROI feature extractor and subsequently decoupled for localization and classification tasks, respectively. For the localization branch, standard fine-tuning strategies are adopted during training. As for the classification branch, we propose the ISCT module to facilitate feature transfer. It fully explores the semantic correlations between base and novel classes and introduces knowledge distillation technology for transfer, thereby helping the novel class learn more precise features. Furthermore, we proposed the SSCL module, which uses the pretrained CLIP [55] text encoder to extract semantic prior knowledge from the class names and subsequently performs supervised contrastive learning under the guidance of semantic prior knowledge to enhance the discriminability of features. A detailed description of the ISCT and SSCL modules will be presented in Sections III-C and III-D.

C. Interclass Semantic Correlation Transfer

In few-shot learning [14], [56], the class prototype (i.e., the mean representation of sample features in the feature space for a given class) can effectively capture the global semantic information of that category. This concept has been successfully incorporated into numerous FSOD algorithms [40], [57], [58], [59]. However, the computational cost of calculating class prototypes using all training samples in each iteration is prohibitively high. To this end, we choose to compute class prototypes based on a feature memory bank. Mathematically, we define the feature memory bank as $M \in \mathbb{R}^{C \times L \times d}$, where C denotes the total number of classes in $C_{\text{base}} \cup C_{\text{novel}}$, L represents the number of stored features per category, and d indicates the feature dimension. Due to the complex background characteristics of remote sensing images, the class prototypes calculated using traditional memory bank update strategies may significantly deviate from the true class prototypes. To obtain more precise class prototypes, we propose a simple yet effective memory bank update strategy, which consists of the following steps.

1) During the first N iterations, the feature memory bank is updated in a first-in–first-out queue manner. This design is adopted because the model's performance remains unstable in early training stages.
2) During the first N iterations, the feature memory bank is updated in a first-in–first-out queue manner. This design is adopted because the model's performance remains unstable in early training stages.
3) Starting from the Nth iteration, features with confidence scores exceeding a predefined threshold t_1 are selectively stored in the memory bank. When the memory bank reaches capacity, the oldest features are overwritten. This design ensures the retention of only high-confidence features, which typically contain purer foreground information and exhibit more precise spatial localization compared to their low-confidence counterparts.
4) Beginning at the Nth iteration, we implement a maximum similarity threshold t_2 to filter out features whose similarity to the current class prototype exceeds t_2. Only features with similarity below t_2 are stored in the memory bank. When the memory bank reaches capacity, the oldest features are overwritten. This design is motivated by the consideration that when new features exhibit excessively high similarity to existing class prototypes, their repeated incorporation would cause the class prototypes to drift toward local optima. The introduced threshold mitigates the risk of prototypes being dominated by a limited number of similar samples.

Based on the above feature memory bank, the mean of the corresponding sample features can be computed as the class prototype. However, due to the limited number of annotated samples for novel classes, the class prototypes computed using

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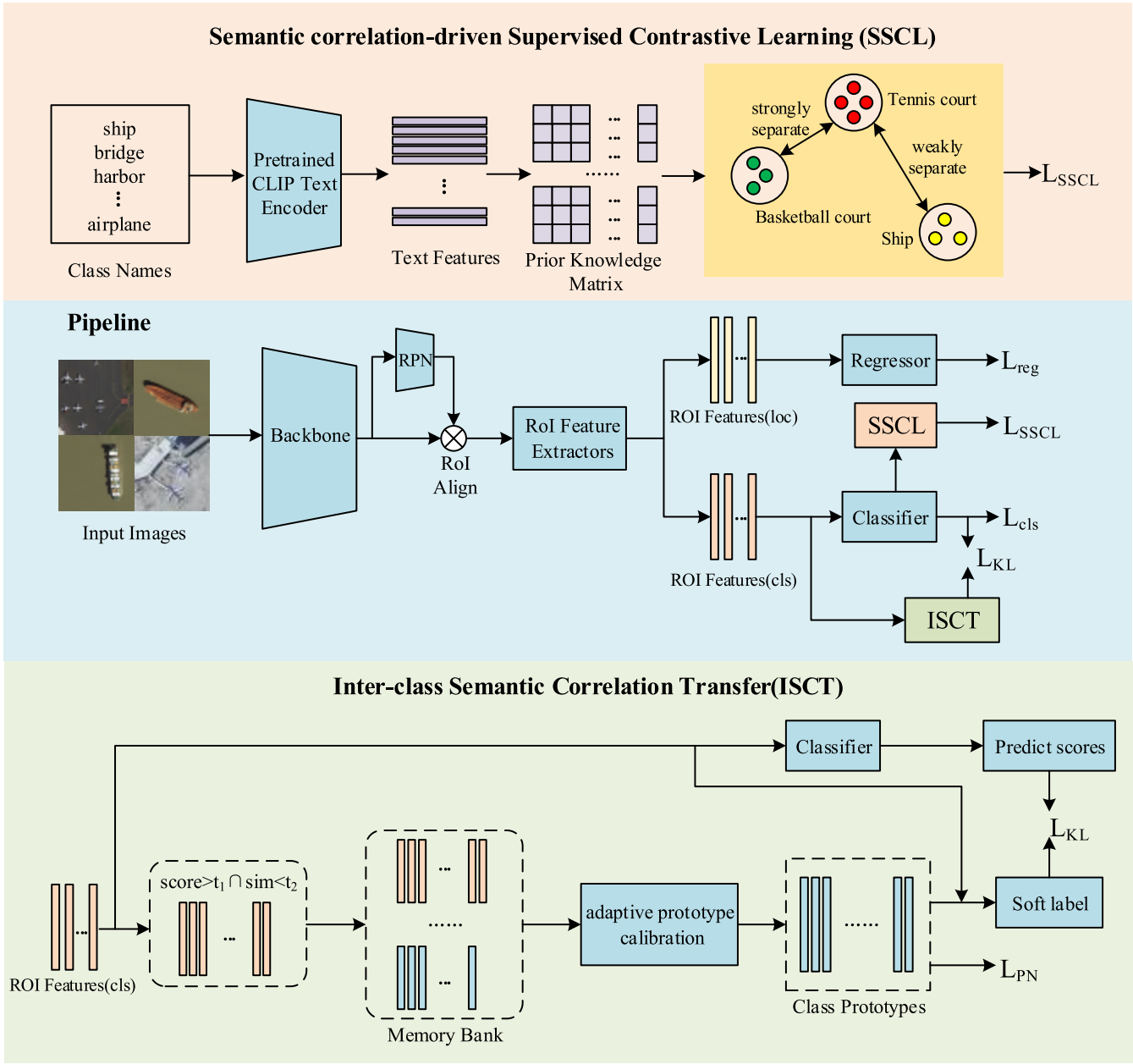


Fig. 2. Overall structure of our proposed SCG-FSOD method. Our method consists of two key components: the ISCT module and the SSCL module. The ISCT module takes the decoupled ROI features for classification as input. First, a memory bank update strategy selects qualified features for storage in the feature memory bank. Then, an adaptive prototype calibration module generates refined class prototypes for novel categories. Next, the ProtoNegative loss function enhances the discriminability of each class prototype. Subsequently, the similarity between ROI features and class prototypes serves as soft labels, enabling ISCT through knowledge distillation. The SSCL module leverages the pretrained CLIP text encoder to extract semantic prior knowledge from remote sensing image class names. This prior knowledge is then utilized to weight negative pairs in supervised contrastive learning, imposing stronger separation constraints on pairs from highly similar categories within the feature space to enhance feature discriminability.

the above method may deviate from the true class centers. To this end, we propose an adaptive prototype calibration module to facilitate more precise class prototype learning for novel classes. Specifically, we calibrate the prototypes of novel classes by leveraging the semantic correlation degrees between novel and base classes. Initially, the mean of the corresponding sample features in the feature memory bank is computed as the initial class prototype. To adaptively measure the semantic correlations between novel and base classes, we introduce the cross-attention mechanism to compute the semantic correlation weights between each novel class and all base classes. These weights are then employed as coefficients to multiply with the base class prototypes, resulting in a weighted summation of the base class prototypes. The corresponding calculation process is as follows:

$$A_i = \text{Softmax}\left(\frac{Q \cdot R^{\mathrm{T}}}{\sqrt{d_k}}\right) \tag{1}$$

$$p_{(w,i)} = A_i V \tag{2}$$

$$Q = p_{(n,i)} W_Q, R = P_b W_R, V = P_b W_V \tag{3}$$

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where P_b denotes the set of all base class prototypes, $p_{(n,i)}$ represents the ith initial novel class prototype, with $\{W_Q, W_R, W_V\}$ as learnable parameter matrices, and d_k is the prototype vector dimension. Here, A_i measures the semantic correlation between $p_{(n,i)}$ and P_b, and $p_{(w,i)}$ is computed as the weighted sum of base class prototypes for $p_{(n,i)}$.

Based on the obtained weighted combination of base class prototypes, the calibrated novel class prototype is derived through the following computational procedure:

$$\tilde{p}_{(n,i)} = \frac{p_{(w,i)} + p_{(n,i)}}{2}. \tag{4}$$

After obtaining the refined class prototypes, we design a ProtoNegative loss to enhance their discriminative ability by treating all pairs of distinct class prototypes as negative pairs and pushing them apart. The mathematical formulation is given as follows:

$$L_{PN} = \frac{1}{N_p} \sum_{i=1}^{N_p} -\log \frac{1}{1 + \sum_{j=1}^{N_p} \mathbb{I}_{i \neq j} \exp(\cos(p_i, p_j)/\tau)} \tag{5}$$

where N_p denotes the number of prototypes, τ represents the temperature hyperparameter, p_i and p_j indicate the ith and jth prototype vectors, respectively, $\cos(p_i, p_j)$ measures the cosine distance between p_i and p_j, and $\mathbb{I}$ is an indicator function that equals 1 when the specified condition is met and 0 otherwise.

After obtaining discriminative class prototypes, we leverage the similarity between the ROI feature and all class prototypes as soft labels, while incorporating knowledge distillation to facilitate the transfer of interclass semantic correlation knowledge. First, we compute the cosine similarity between ROI feature f_r and class prototypes p_i as follows:

$$s_r^i = \eta \cdot \frac{f_r \cdot p_i}{\|f_r\| \|p_i\|} \tag{6}$$

where η is the scaling factor. To better extract semantic correlation between base and novel classes, we follow MFDC [60] by setting the similarity between ROI features and other base class prototypes to a small constant value, as follows:

$$s_r^j = -\eta, j \in C_{\text{base}} \setminus \{c_{\text{gt}}\} \tag{7}$$

where η is the same scaling factor as in (6). Subsequently, we normalize the similarities between the ROI feature f_r and all class prototypes using the softmax function, as follows:

$$q_r^c = \frac{\exp(s_r^i)}{\sum_{i=1}^{C} \exp(s_r^i)}, c \in C_{\text{novel}} \cup C_{\text{base}} \tag{8}$$

where C denotes the total number of class prototypes. In addition, we set the similarity between each feature f_r and the background c_{bg} to zero, thereby obtaining its corresponding soft label $q_r = [q_r, 0]$.

Finally, to perform knowledge distillation, we treat q_r as the teacher and the classifier's predicted vector z_r as the student, minimizing the Kullback–Leibler (KL) divergence between them as follows:

$$L_{\text{KL}} = \sum_{i=1}^{N_r} q_r \log\left(\frac{q_r}{z_r}\right) \tag{9}$$

where N_r denotes the number of foreground ROI features.

D. Semantic Correlation-Driven Supervised Contrastive Learning

Remote sensing images exhibit high interclass similarity and substantial intraclass diversity, which exacerbates classification confusion in FSOD. To learn more discriminative features, we incorporate supervised contrastive learning into the classifier. In contrast to self-supervised contrastive learning, supervised contrastive learning explicitly leverages label semantic information to treat features from the same class as positive pairs and pull them closer in the feature space, while pushing apart those from different classes as negative pairs. The corresponding loss function can be formulated as follows:

$$L_{\text{SCL}} = -\frac{1}{N} \sum_{i=1}^{N} -\log \frac{\sum_{j=1, j\neq i}^{N} \mathbb{I}_{y_j = y_i} \exp(f_i \cdot f_j/\tau)}{\sum_{j=1, j\neq i}^{N} \exp(f_i \cdot f_j/\tau)} \tag{10}$$

where f denotes the feature vector of foreground objects in the classifier, N represents the total number of features, τ is the temperature hyperparameter, and y_i and y_j denote the category labels of f_i and f_j, respectively. $\mathbb{I}(\cdot)$ is the indicator function that outputs 1 if the specified condition is satisfied and 0 otherwise.

In supervised contrastive learning, all negative pairs are typically weighted equally. However, the actual dissimilarity between negative pairs varies significantly. As illustrated in Fig. 1, the tennis court and ship exhibit substantial differences, making them unlikely to be confused during fine-tuning. Thus, their feature representations should be weakly separated in the supervised contrastive learning framework. Conversely, the tennis court and basketball court share high visual similarity, which may lead to misclassification during fine-tuning. To mitigate this, their representations should be strongly separated in supervised contrastive learning.

To address this issue, we consider leveraging the semantic correlations between categories to assign weights to negative samples. Building upon the remarkable zero-shot learning capabilities of large-scale pretrained models, particularly CLIP, we decided to extract text embeddings from remote sensing image class names using CLIP's text encoder. These embeddings are subsequently employed to construct a prior knowledge matrix for weighting negative samples in supervised contrastive learning. Specifically, given a class name (e.g., ship), a prompt is generated using a predefined prompt template "a centered satellite photograph of {class name}." This prompt is then fed into the pretrained CLIP text encoder to obtain the corresponding text feature representation f^{text}. Subsequently, the similarity $S^{\text{text}}_{(c_1,c_2)}$ between two categories c_1 and c_2 can be expressed as follows:

$$S^{\text{text}}_{(c_1,c_2)} = \frac{f^{\text{text}}_{c_1} \cdot f^{\text{text}}_{c_2}}{\|f^{\text{text}}_{c_1}\| \|f^{\text{text}}_{c_2}\|}. \tag{11}$$

Based on the intercategory similarity described above, we construct a prior knowledge matrix φ. For C foreground object categories, the prior knowledge matrix can be formulated as $\varphi \in \mathbb{R}^{C\times C}$. For any pair of categories (c_1, c_2), where $1 \leq c_1$ and $c_2 \leq C$, we have $\varphi_{(c_1,c_2)} = S^{\text{text}}_{(c_1,c_2)}$. Notably, φ is a symmetric matrix with the main diagonal value being 1. Subsequently, we leverage the prior semantic correlations encoded in φ

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TABLE I
FOUR DISTINCT NOVEL/BASE CLASS SPLIT SETTINGS ON THE DIOR DATASET

Split	Novel classes					Base classes
1	Baseball field	Basketball court	Bridge	Chimney	Ship	Rest
2	Airplane	Airport	Expressway toll station	Harbor	Ground track field	Rest
3	Dam	Golf course	Storage tank	Tennis court	Vehicle	Rest
4	Expressway service area	Overpass	Stadium	Train station	Windmill	Rest

to reweight negative sample pairs in supervised contrastive learning. The corresponding loss function can be expressed as

$$L_{\text{SSCL}} = -\frac{1}{N}\sum_{i=1}^{N} -\log \frac{\sum_{j=1,j\neq i}^{N} \mathbb{I}_{y_j=y_i} \exp(f_i \cdot f_j/\tau)}{\sum_{j=1,j\neq i}^{N} \exp(\varphi_{(y_i,y_j)} f_i \cdot f_j/\tau)}. \quad (12)$$

This enhances the weight of negative samples between similar categories, thereby more effectively addressing the classification confusion issue.

E. Loss Function

During the base training phase, we employ the standard Faster R-CNN [54] loss function to train the model, formulated as follows:

$$L_{\text{det}} = L_{\text{rpn}} + L_{\text{cls}} + L_{\text{reg}} \quad (13)$$

where L_{rpn} denotes the RPN loss that distinguishes foreground/background proposals and performs coarse bounding-box regression, L_{cls} represents the cross-entropy classification loss, and L_{reg} corresponds to the smooth $L1$ regression loss for bounding-box refinement. In the few-shot fine-tuning phase, the joint loss function to be optimized is formulated as

$$L = L_{\text{det}} + \alpha L_{\text{pn}} + \beta L_{\text{KL}} + \lambda L_{\text{SSCL}} \quad (14)$$

where α, β, and λ are the hyperparameters that balance the contributions of different loss terms.

IV. EXPERIMENT

A. Datasets and Evaluation Metrics

To validate the effectiveness of our proposed method, we conducted extensive experiments on two datasets: DIOR and NWPU VHR-10.v2.

1) *DIOR [35]:* The DIOR dataset is a large-scale benchmark dataset designed for object detection in optical remote sensing images. It comprises 23 463 remote sensing images and 1 92 472 object instances, covering 20 common object categories. Each image has a uniform size of 800 × 800 pixels, with spatial resolution ranging from 0.5 to 30 m. The dataset is partitioned into training set (5862 images), validation set (5863 images), and testing set (11 738 images). Following the FSOD settings in [40] and [41], we divided the DIOR dataset into four distinct novel/base class splits. As shown in Table I, each split contains five novel classes and 15 base classes, with no overlapping novel classes across different splits.
2) *NWPU VHR-10.v2 [36]:* NWPU VHR-10.v2 is a classic high-resolution remote sensing image object detection dataset comprising ten categories of geospatial objects. The dataset consists of 1172 images uniformly sized at 400 × 400 pixels, partitioned into training set (679 images), validation set (200 images), and testing set (293 images). Following the novel/base class split in [41] and [43], each split consists of seven base classes and three novel classes, as specified in Table II.

TABLE II
TWO DIFFERENT NOVEL/BASE CLASS SPLIT SETTINGS ON THE NWPU VHR-10.V2 DATASET

Split	Novel classes			Base classes
1	Airplane	Baseball diamond	Tennis court	Rest
2	Basketball	Ground track field	Vehicle	Rest

3) *Evaluation Metric:* Following previous study [40], [41], [43], we employ the mean average precision (mAP) at an intersection over union (IoU) threshold of 0.5 as the evaluation metric. The mAP is calculated by averaging the average precision (AP) values across all categories. Typically, AP is computed as follows:

$$\text{AP} = \sum_{R\in(0,0.1,\ldots,1)} \frac{1}{11} P(R) \quad (15)$$

where $P(R)$ denotes the precision–recall curve.

B. Implementation Details

In this article, all experiments were conducted utilizing the PyTorch [61] framework on a computational platform equipped with NVIDIA GeForce RTX 3090 GPU. Specifically, our proposed SCG-FSOD is built upon the Faster R-CNN [54] framework, utilizing the ResNet-101 [10] pretrained on ImageNet [62] as the backbone. During the base training phase, the model was trained for a total of 15 000 iterations. The learning rate was initialized at 0.02 after warm-up, and then reduced by a factor of 10 at 10 000 and 13 300 iterations. During the few-shot fine-tuning stage, the model was trained for 3K iterations with an initial learning rate of 0.01. We fine-tuned the detector using samples from all categories, randomly selecting K samples per class with K set to 3, 5, 10, and 20, respectively. In the above two stages, we employed stochastic gradient descent (SGD) as the optimizer with a momentum coefficient of 0.9 and a weight decay of $5e^{-5}$. Furthermore, the values of L and d in the memory bank were both set to 2048. In the memory update strategy, the values of N, t_1, and t_2 were configured as 1000, 0.5, and 0.85, respectively. The

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TABLE III

EXPERIMENTAL RESULTS OF DIFFERENT FSOD METHODS ON THE DIOR DATASET UNDER THE K-SHOT (K = 3, 5, 10, 20) TESTING SETTINGS. ALL REPORTED RESULTS REPRESENT THE MAP (%) FOR NOVEL CLASSES. THE BEST PERFORMANCE IS HIGHLIGHTED IN RED, AND THE SECOND-BEST IS MARKED IN BLUE. "AVG." DENOTES THE AVERAGE RESULTS OVER THE FOUR SPLITS

Split	Shot	P-CNN [40]	SAGS [63]	MSOCL [53]	G-FSDet [41]	AMTN [64]	SAE-FSDet [65]	ST-FSOD [66]	GE-FSOD [44]	RIFR [32]	Ours
Split1	3-shot	18.00	29.30	28.50	27.57	22.00	28.80	27.80	31.69	28.51	32.99 (+1.30)
	5-shot	22.80	31.60	34.60	30.52	24.71	32.40	31.23	34.88	32.34	34.35 (-)
	10-shot	27.60	36.40	40.80	37.46	29.50	37.09	37.59	38.02	37.74	39.84 (-)
	20-shot	29.60	40.20	43.30	39.83	32.20	42.46	39.26	43.08	41.32	43.63 (+0.33)
Split2	3-shot	14.50	12.60	16.80	14.13	15.10	13.99	14.61	14.48	15.11	18.31 (+1.51)
	5-shot	14.90	15.50	20.60	15.84	18.10	15.65	16.94	16.68	18.75	23.35 (+2.75)
	10-shot	18.90	19.70	23.60	20.70	20.60	17.41	19.93	19.08	21.93	28.41 (+4.81)
	20-shot	22.80	23.80	29.20	22.69	24.70	21.34	23.03	26.16	24.10	34.25 (+5.05)
Split3	3-shot	16.50	20.90	19.60	16.03	18.50	16.74	17.77	18.85	17.78	20.09 (-)
	5-shot	18.80	24.80	21.40	23.25	20.50	19.07	24.27	22.58	24.15	24.34 (-)
	10-shot	23.30	28.60	26.00	26.24	23.70	28.44	27.07	30.46	26.46	33.23 (+2.77)
	20-shot	28.80	36.10	30.20	32.05	29.20	29.88	33.49	31.32	29.76	36.97 (+0.87)
Split4	3-shot	15.20	17.50	15.60	16.74	/	17.27	17.34	20.66	18.22	18.20 (-)
	5-shot	17.50	19.70	19.00	21.03	/	20.48	21.70	22.45	22.87	24.19 (+1.32)
	10-shot	18.90	23.70	24.60	25.84	/	22.69	26.45	26.06	28.49	30.80 (+2.31)
	20-shot	25.70	27.70	29.40	31.78	/	26.75	31.80	29.47	32.12	36.83 (+4.71)
Avg.	3-shot	16.05	20.08	20.13	18.62	18.53	19.20	19.38	21.42	19.91	22.40 (+0.98)
	5-shot	18.50	22.90	23.90	22.66	21.10	21.90	23.54	24.15	24.53	26.56 (+2.03)
	10-shot	22.18	27.10	28.75	27.56	24.60	26.41	27.76	28.41	28.66	33.07 (+4.32)
	20-shot	26.73	31.95	33.03	31.59	28.70	30.11	31.90	32.51	31.83	37.92 (+4.89)

TABLE IV

EXPERIMENTAL RESULTS OF DIFFERENT FSOD METHODS ON THE NWPU VHR-10.V2 DATASET UNDER K-SHOT (K = 3, 5, 10, 20) TESTING SETTINGS. ALL REPORTED RESULTS REPRESENT THE MAP (%) FOR NOVEL CLASSES. THE BEST PERFORMANCE IS HIGHLIGHTED IN RED, AND THE SECOND-BEST IS MARKED IN BLUE. "AVG." DENOTES THE AVERAGE RESULTS OVER THE TWO SPLITS

Split	Shot	FsDetView [67]	P-CNN [40]	DeFRCN [28]	G-FSDet [41]	Incremental-DETR [68]	TEMO [69]	RP-FSOD [43]	RIFR [32]	Ours
Split1	3-shot	24.50	41.80	45.59	49.05	50.90	57.50	50.45	53.17	51.52 (-)
	5-shot	29.50	49.17	52.62	56.10	53.10	66.70	60.72	62.39	64.95 (-)
	10-shot	31.70	63.29	63.46	71.82	69.90	75.00	72.91	72.79	75.99 (+0.99)
	20-shot	32.70	66.83	70.88	75.41	73.10	75.80	81.46	79.46	82.21 (+0.75)
Split2	3-shot	39.00	39.32	26.03	50.09	47.30	50.80	51.91	/	53.54 (+1.63)
	5-shot	40.30	46.10	37.14	58.75	57.20	59.30	59.59	/	64.33 (+4.74)
	10-shot	45.00	55.90	45.77	67.00	65.60	64.50	69.22	/	71.94 (+2.72)
	20-shot	46.20	58.37	58.04	75.86	70.50	73.30	79.24	/	81.99 (+2.75)
Avg.	3-shot	31.75	40.56	35.81	49.57	49.10	54.15	51.18	53.17	52.53 (-)
	5-shot	34.90	47.64	44.88	57.43	55.15	63.00	60.16	62.39	64.64 (+1.64)
	10-shot	38.35	59.60	54.62	69.41	67.75	69.75	71.07	72.79	73.97 (+1.18)
	20-shot	39.45	62.60	64.46	75.64	71.80	74.55	80.35	79.46	82.10 (+1.75)

scaling factor η was set to 10, and the temperature parameter τ was set to 0.1.

C. Experimental Results and Comparisons

To demonstrate the effectiveness of our proposed SCG-FSOD, we compared its performance with several state-of-the-art methods on the DIOR and NWPU VHR-10.v2 datasets. The compared FSOD methods include: P-CNN [40], SAGS [63], MSOCL [53], G-FSDet [41], AMTN [64], SAE-FSDet [65], ST-FSOD [66], GE-FSOD [44], RIFR [32], FsDetView [67], DeFRCN [28], Incremental-DETR [68], TEMO [69], and RP-FSOD [43]. The experimental results are summarized in Tables III and IV.

1) Experimental Results on DIOR Dataset: Table III presents the experimental results of our method compared with several state-of-the-art approaches on the DIOR dataset.

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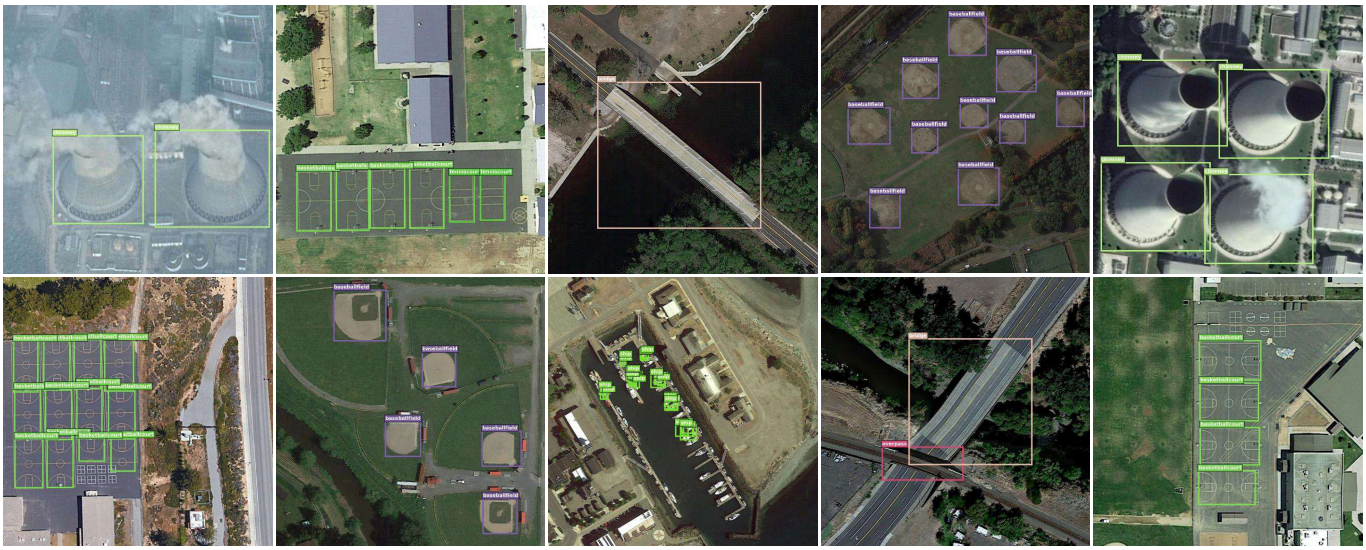


Fig. 3. Visualization of detection results on the DIOR test set in the first novel/base split under the ten-shot setting. Novel classes: baseball field, basketball court, bridge, chimney, and ship.

It can be clearly observed that our method demonstrates overall superior performance compared to other methods. Specifically, our method achieves the best performance in 11 out of 16 experimental settings and obtains the second-best results in three settings. For instance, in split1, our method reaches 32.99% and 43.63% performance under three-shot and 20-shot settings, surpassing the best-performing comparative methods, GE-FSOD and MSOCL, by margins of 1.30% and 0.33%, respectively. Furthermore, our method achieves the second-best performance under the ten-shot setting while demonstrating only marginal performance gaps compared to the top-performing method, GE-FSOD, and the runner-up method MSOCL in the five-shot configuration. In split2, our method achieves the best performance in all four experimental settings. This is the split where our method performs best. In split3, our method achieved superior performance with 33.23% and 36.97% under the ten-shot and 20-shot settings, surpassing the strongest competitors, GE-FSOD and SAGS, by margins of 2.77% and 0.87%, respectively. Meanwhile, it attained the second-best results under the three- and five-shot configuration. Finally, in split4, our method achieved the best performance across all five-, ten-, and 20-shot settings, reaching 24.19%, 30.80%, and 36.83%, respectively, and surpassing the second-best method by margins of 1.32%, 2.31%, and 4.71%. These encouraging experimental results compellingly demonstrate the effectiveness of our proposed SCG-FSOD method.

Furthermore, the varying difficulty levels of novel classes across different dataset splits can lead to fluctuations in model performance. To this end, we report the average results of different methods over four splits. It can be observed that our method achieved the best average performance across all four splits. For example, in the five-shot setting, our method achieves a 2.03% improvement over the best-performing comparison algorithm, RIFR. Furthermore, under the ten-shot configuration, the enhancement relative to the strongest competitor MSOCL is more pronounced, reaching an encouraging 4.32%. These superior experimental results further demonstrate the effectiveness of our method.

In addition to the quantitative analysis presented above, we visualize the detection results on the DIOR test set under the ten-shot setting for the first novel/base split, as illustrated in Fig. 3. It can be clearly observed that our model effectively detects the majority of novel objects, demonstrating superior generalization performance.

2) Experimental Results on NWPU VHR-10.v2 Dataset: Compared to the DIOR dataset, the NWPU VHR-10.v2 dataset contains fewer categories, and the number of images and objects per category is relatively limited, which poses a more significant challenge for FSOD tasks. Table IV presents the experimental results of our method compared with several state-of-the-art approaches on the NWPU VHR-10.v2 dataset. It can be observed that our method significantly outperforms other compared methods. Specifically, our method achieves the best performance in six out of the eight experimental settings and obtains the second-best result in one setting. For instance, in split1, our method achieves performance of 75.99% and 82.21% under the ten-shot and 20-shot settings, outperforming the strongest competitors TEMO and RP-FSOD by 0.99% and 0.75%, respectively. However, it attains the second-best performance under the five-shot setting. In split 2, our method achieves the best performance across all four experimental settings, outperforming RP-FSOD, which consistently ranks second in all four settings, by margins of 1.63%, 4.74%, 2.72%, and 2.75%, respectively. These promising experimental results demonstrate the effectiveness of the proposed SCG-FSOD method. Consistent with the evaluation on the DIOR dataset, we also report the average results of different methods across two splits. It can be observed that our method achieves the best overall performance under the five-shot, ten-shot, and 20-shot settings. For example, under the five-shot setting, our method achieves an average performance of

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Fig. 4. Visualization of detection results on the NWPU VHR-10.v2 test set in the first novel/base split under the ten-shot setting. Novel classes: airplane, baseball diamond, and tennis court.

TABLE V

ABLATION STUDY FOR EACH COMPONENT OF SCG-FSOD. ALL REPORTED RESULTS REPRESENT THE MAP (%) FOR NOVEL CLASSES. THE BEST PERFORMANCE IS HIGHLIGHTED IN RED

	Baseline	ISCT	SSCL	3-shot	5-shot	10-shot	20-shot
I	√			24.38	26.87	32.26	36.50
II	√	√		28.44	31.67	36.24	41.60
III	√		√	29.46	29.76	36.99	40.21
IV	√	√	√	32.99	34.35	39.84	43.63

64.64%, outperforming the second-best method TEMO by a margin of 1.64%. In the ten-shot configuration, it surpasses the strongest competitor RIFS with an improvement of 1.18%. These excellent experimental results further demonstrate the efficacy of our proposed algorithm.

In addition, the detection results for the NWPU VHR-10.v2 test set in the first novel/base split under the ten-shot setting are visualized in Fig. 4. It can be observed that our model accurately detects novel class objects, exhibiting excellent performance.

D. Ablation Study

To validate the effectiveness of each module in our proposed SCG-FSOD, extensive ablation experiments were conducted. All experiments in this section were performed on the DIOR test set in the first novel/base split.

1) Ablation Study for Each Component of SCG-FSOD: The proposed SCG-FSOD method comprises two key components: the ISCT module and the SSCL module. To demonstrate the effectiveness of these modules, ablation studies were conducted, and their performance is summarized in Table V. It can be clearly observed that both the ISCT and SSCL modules contribute to a consistent improvement in model performance. Specifically, we introduce the ISCT module upon a baseline method. It sufficiently explores semantic correlations between base and novel classes and incorporates knowledge distillation for transfer, thereby mitigating base-class bias and improving the discriminability of novel-class features. By comparing rows I and II in Table V, it can be observed that the ISCT module achieves performance improvements of 4.06%, 4.80%, 3.98%, and 5.10% under the three-shot, five-shot, ten-shot, and 20-shot settings, respectively. Furthermore, to address the issue that interclass similarity and intraclass diversity in remote sensing images exacerbate classification confusion, we propose the SSCL module. This module leverages prior knowledge to weight negative pairs in supervised contrastive learning, imposing stronger separation constraints on negative pairs of highly similar categories in the feature space, thereby effectively mitigating classification confusion. By comparing rows I and III in Table V, it is evident that the SSCL module yields performance gains of 5.08%, 2.89%, 4.73%, and 3.71% under the three-shot, five-shot, ten-shot, and 20-shot settings, respectively. Finally, by integrating both the ISCT and SSCL modules, the model achieves superior performance, surpassing the baseline method by 8.61%, 7.48%, 7.58%, and 7.13% under the three-shot, five-shot, ten-shot, and 20-shot settings, respectively.

In addition to the quantitative analysis above, we have also visualized the detection results of the four models, as shown in Fig. 5. It can be clearly observed that the SCG-FSOD model, which integrates the ISCT and SSCL modules, detects various novel categories more accurately than the baseline model, demonstrating superior performance.

2) Ablation Study on ISCT Moudle: As presented in Table VI, we conducted an ablation study on the proposed ISCT module. This study was designed to evaluate the effectiveness of its three key components: the memory bank update strategy, the adaptive prototype calibration module, and the ProtoNegative loss. It can be clearly observed from Table VI that the memory bank update strategy, the adaptive prototype

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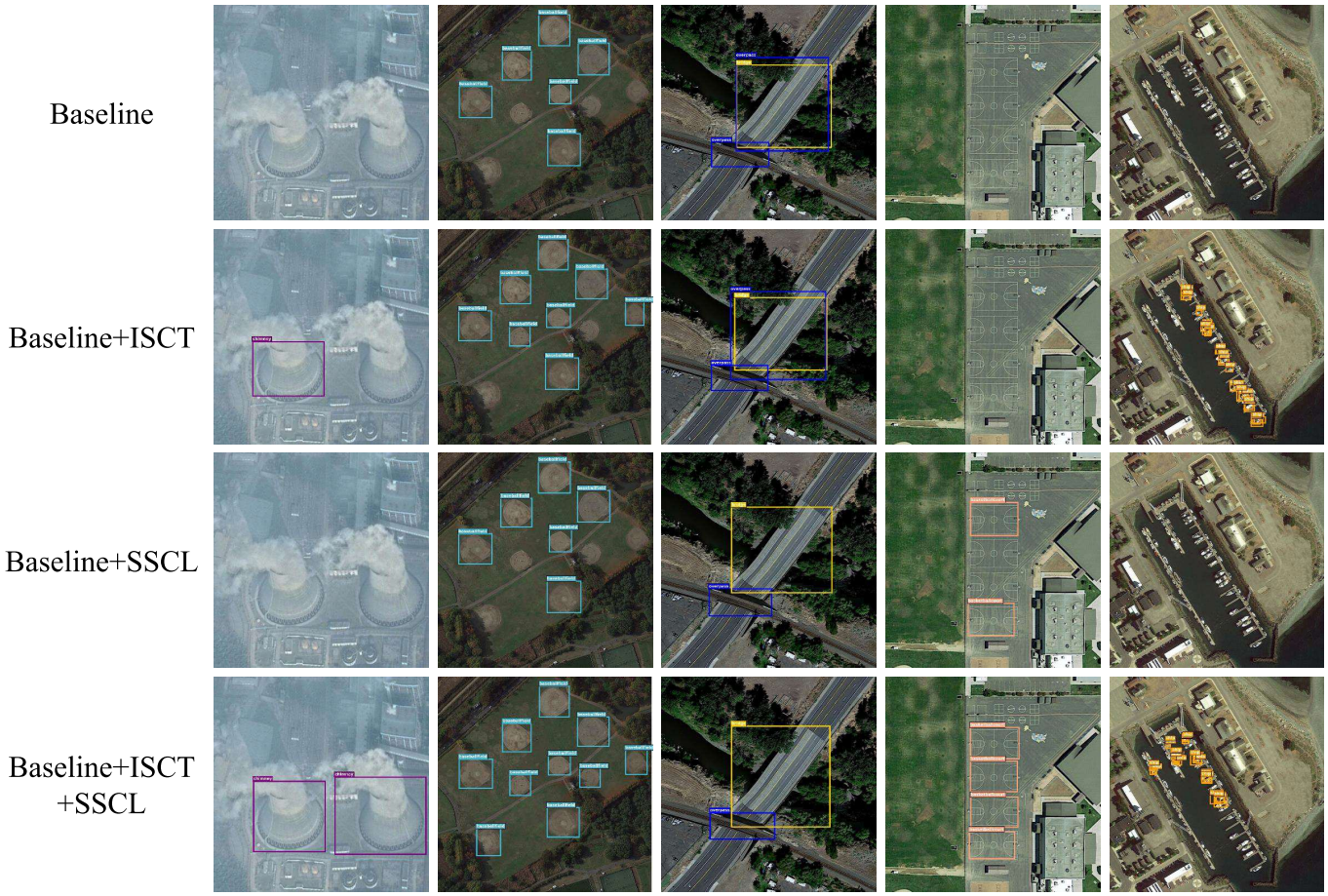


Fig. 5. Visualization of ablation study results for each component of SCG-FSOD. Experiments were conducted on the first novel/base split of the DIOR test set under the ten-shot setting. Novel classes: baseball field, basketball court, bridge, chimney, and ship.

TABLE VI

ABLATION STUDY ON ISCT MODULE. ALL REPORTED RESULTS REPRESENT THE MAP (%) FOR NOVEL CLASSES. THE BEST PERFORMANCE IS HIGHLIGHTED IN RED. THE TERMS "SCG-FSOD-MBUS," "SCG-FSOD-APCM," AND "SCG-FSOD-PNLOSS" DENOTE THE SCG-FSOD FRAMEWORK WITHOUT THE MEMORY BANK UPDATE STRATEGY, WITHOUT THE ADAPTIVE PROTOTYPE CALIBRATION MODULE, AND WITHOUT THE PROTONEGATIVE LOSS, RESPECTIVELY

Method	3-shot	5-shot	10-shot	20-shot
SCG-FSOD - MBUS	31.88	32.94	39.26	42.71
SCG-FSOD - APCM	32.13	33.22	38.32	42.14
SCG-FSOD - PNloss	31.67	33.27	39.37	42.66
SCG-FSOD	32.99	34.35	39.84	43.63

TABLE VII

ABLATION STUDY OF PRIOR KNOWLEDGE MATRIX IN SSCL MODULE. ALL REPORTED RESULTS REPRESENT THE MAP (%) FOR NOVEL CLASSES. THE BEST PERFORMANCE IS HIGHLIGHTED IN RED. THE TERM "SCG-FSOD-PKM" DENOTES SCG-FSOD WITHOUT THE PRIOR KNOWLEDGE MATRIX

Method	3-shot	5-shot	10-shot	20-shot
SCG-FSOD - PKM	32.46	33.13	39.69	42.42
SCG-FSOD	32.99	34.35	39.84	43.63

calibration module, and the ProtoNegative loss all lead to performance improvements across all four experimental settings. Specifically, the memory bank update strategy enhances the precision of class prototypes by selecting high-confidence features and excluding redundant ones. This strategy led to performance gains of 1.11%, 1.41%, 0.58%, and 0.92% under the four experimental settings, respectively. The adaptive prototype calibration module leverages the semantic correlation between novel and base classes to calibrate the prototypes for novel classes, which resulted in enhancements of 0.86%, 1.13%, 1.52%, and 1.49% under the three-shot, five-shot, ten-shot, and 20-shot settings, respectively. The ProtoNegative loss is designed to enhance the discriminative ability of each class prototype, achieving performance improvements of 1.32%, 1.08%, 0.47%, and 0.97% under the four experimental settings, respectively. The encouraging results described above serve as strong evidence for the efficacy of the memory bank update strategy, the adaptive prototype calibration module, and the ProtoNegative loss.

3) Ablation Study of Prior Knowledge Matrix in SSCL Module: We performed ablation studies on the prior knowledge matrix in the SSCL module, as summarized in Table VII. It can be intuitively observed that the prior knowledge matrix

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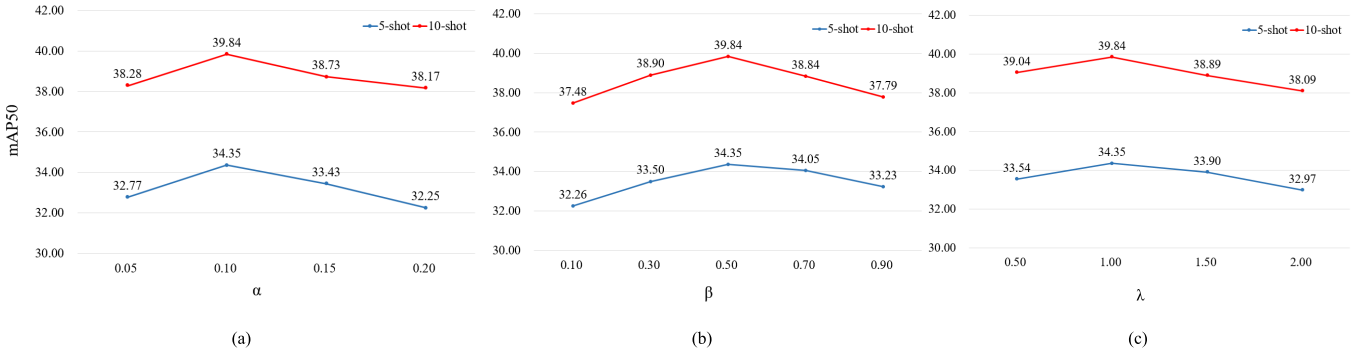


Fig. 6. Ablation study on the parameters of the loss function. (a) Impact of parameter α on performance. (b) Impact of parameter β on performance. (c) Impact of parameter γ on performance.

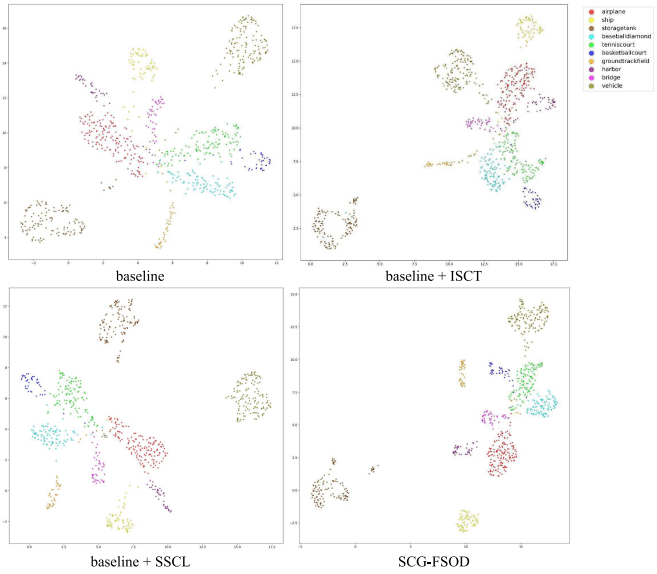


Fig. 7. UMAP visualization of feature embeddings from the NWPU VHR-10.v2 test. Novel classes: airplane, baseball diamond, and tennis court.

enhances model performance across all four experimental settings. Specifically, the prior knowledge matrix weights negative pairs in supervised contrastive learning to enforce a stronger separation constraint on highly similar category pairs in the feature space. Under this constraint, the detector achieves improved discrimination between different classes of objects. Under the four experimental settings, the incorporation of the prior knowledge matrix led to performance gains of 0.53%, 1.22%, 0.15%, and 1.21%, respectively. These promising results demonstrate the effectiveness of the prior knowledge matrix.

4) Ablation Study on the Parameters of the Loss Function: As described in Section III-E, the hyperparameters α, β, and λ are used to balance the contributions of different loss terms in the joint loss function during the few-shot fine-tuning stage. As shown in Fig. 6, we analyze the effects of hyperparameters α, β, and λ under both five-shot and ten-shot settings. As illustrated in Fig. 6(a), by varying α within the range of [0.05, 0.20], it can be clearly observed that the proposed SCG-FSOD method achieves optimal performance under both settings when α is set to 0.10. Subsequently, the performance of SCG-FSOD declines as α increases. As shown in Fig. 6(b), when β is varied within the range of [0.10, 0.90], it can be observed that the performance of SCG-FSOD gradually improves with increasing β, reaching its peak under both five-shot and ten-shot settings when β is set to 0.50. As presented in Fig. 6(c), when λ is varied within the range of [0.50, 2.00], it can be clearly observed that the performance of SCG-FSOD achieves its optimum under both settings when λ is set to 1.00. However, as λ increases beyond 1.00, the performance gradually decreases. Based on the aforementioned experimental results, the optimal values for the hyperparameters are determined as $\alpha = 0.10$, $\beta = 0.50$, and $\lambda = 1.00$.

E. Further Analysis

1) Analysis of Feature Visualization: To analyze the impact of the ISCT and SSCL modules on feature distribution, we performed experiments on the first novel/base split of the NWPU VHR-10.v2 test set and visualized the resulting feature distributions using UMAP [70], as shown in Fig. 7. As observed from the top of the figure, the feature space of baseline + ISCT exhibits a more distinct decision boundary between the novel class "tennis court" (green dots) and its semantically similar base class "basketball court" (blue dots) compared to that of the baseline. Furthermore, in comparison to baseline + SSCL, the proposed SCG-FSOD, owing to the incorporation of the ISCT module, achieves greater feature distinguishability in its embedding space between the novel classes ["airplane" (red dots), "baseball diamond" (cyan dots), and "tennis court" (green dots)] and the surrounding base classes. These observations confirm the efficacy of the ISCT module in reducing base class bias.

Simultaneously, as shown in the first column of the figure, compared to the baseline, the feature space of baseline + SSCL exhibits an increased distance between the novel class "tennis court" (green dots) and the semantically similar base class "basketball court" (blue dots), resulting in a sharper boundary between the two feature clusters. Furthermore, compared to baseline + ISCT, the proposed SCG-FSOD with the SSCL module exhibits more compact intraclass distributions and more separable interclass distributions in its feature space. These observations demonstrate that the SSCL module effectively increases the feature distance between similar classes.

2) Analysis of Model Efficiency: In addition to analyzing the detection performance, we performed a comparative

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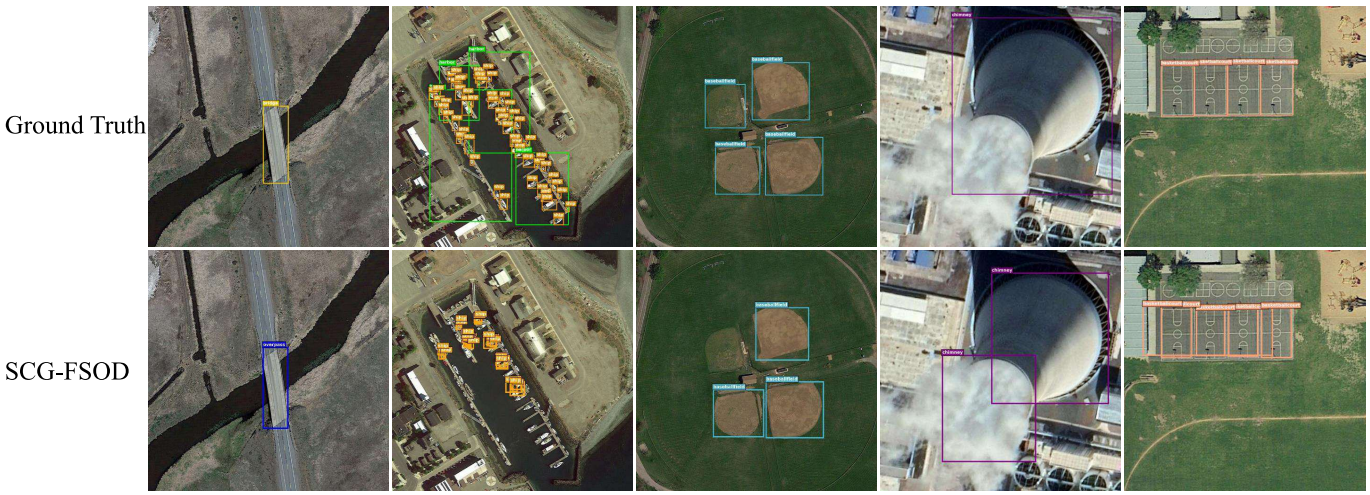


Fig. 8. Visualization of detection failure examples for the SCG-FSOD method on the first novel/base split of the DIOR test set under the ten-shot setting. Novel classes: baseball field, basketball court, bridge, chimney, and ship.

TABLE VIII
COMPARATIVE ANALYSIS OF MODEL EFFICIENCY BETWEEN OUR PROPOSED SCG-FSOD AND OTHER FSOD APPROACHES. THE BEST PERFORMANCE IS HIGHLIGHTED IN RED

Method	P-CNN [40]	SAGS [63]	FsDetView [67]	G-FSDet [41]	RP-FSOD [43]	SAE-FSDet [65]	ST-FSOD [66]	SCG-FSOD
Params.	56.53M	49.58M	73.96M	74.08M	65.76M	60.89M	58.96M	73.1M
GFLOPs	/	327.17	264.56	/	121.1	181.13	464.3	111.18

analysis of model efficiency between SCG-FSOD and other FSOD methods, as shown in Table VIII. As can be observed, compared with FsDetView and G-FSDet, our SCG-FSOD method requires fewer parameters while achieving superior performance. In contrast to methods with even fewer parameters, such as P-CNN and SAGS, SCG-FSOD attains enhanced detection performance with only a marginal increase in model complexity. Although SCG-FSOD is not the method with the fewest parameters, it demonstrates exceptional computational efficiency, achieving the lowest GFLOPs among all compared methods. These comparative results further demonstrate the effectiveness of our proposed method.

3) Limitations: Although our method demonstrates promising performance, it still exhibits certain limitations. To better understand these limitations, we visualize several failure cases of the SCG-FSOD method on the DIOR test set, as illustrated in Fig. 8. As illustrated in the figure, our method exhibits certain issues in these scenarios, including misclassifications, missed detections, and duplicate detected boxes. The performance of our approach is consequently constrained by these limitations. The reasons for the failures in these cases can be roughly categorized into three types.

1) Misclassification caused by high interclass similarity, where objects are mistakenly identified as semantically similar categories (e.g., "bridge" classified as "overpass" in the first column).
2) Missed detections resulting from either small target size (second column) or high similarity between the target and the background (third column), both of which lead to insufficiently salient features for accurate detection.
3) Duplicate detection boxes due to unclear target boundaries, as exemplified in the fourth and fifth columns. Therefore, addressing these issues represents several aspects targeted for improvement in our future work.

V. CONCLUSION

This article proposes a novel FSOD method for remote sensing images, termed SCG-FSOD, which aims to enhance the detection performance for novel classes by leveraging interclass semantic correlations. The framework comprises two key components: ISCT and SSCL modules. The ISCT module first extensively explores semantic correlations between base and novel classes, and subsequently introduces knowledge distillation to guide the detector in learning interclass semantic correlations. This process effectively mitigates the model's base-class bias while enhancing the discriminability of novel-class features. Furthermore, to address the exacerbated classification confusion resulting from high interclass similarity and significant intraclass variance in remote sensing imagery, we design an SSCL module. This module utilizes semantic prior knowledge to guide the contrastive learning process, thereby significantly enhancing the feature discriminability in remote sensing scenarios. Extensive experiments on two benchmark remote sensing FSOD datasets demonstrate the effectiveness of our proposed SCG-FSOD method.

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